

## European Equity Derivatives Outlook

### A Practical Dispersion Trading Handbook

- We update the in-depth studies on dispersion trading published this year and discuss a set of improvements. These studies can also be helpful for index volatility trading, which benefits from better insights into correlation dynamics.
- We discuss the long-term and recent drivers of performance in Euro STOXX 50 and S&P 500 dispersion trades. We focus particularly on the recent challenges for European trades, which in our view are mainly: the deterioration of the **entry level** for the vol spread, the **insufficient notional sold on the index leg** and the volatility impact of the **6 stocks that are generally excluded from the Euro STOXX 50 basket**.
- We identify a set of features that can be leveraged to improve dispersion trading outcomes and partially overcome the current challenges. We summarize our in-depth findings related to earnings seasons, which are crucial periods to trade for dispersion players: we demonstrate that the **single-stock vs index vol spread generally underprices earnings risk premium**, we identify **earnings-sensitive stocks**, we **discuss the trade-off between gamma and vol reset** and potential adjustments based on **seasonality**.
- Finally, we suggest some further techniques that can be applied to dispersion trading such as **overlaying VSTOXX call spread collars**, **“legging” the index volatility execution** and **tactically overlaying reverse dispersion** to improve carry.

#### Global Quantitative and Derivatives Strategy

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13 September 2023

## Recently Published Research

**Table 1: Recently published European trade ideas and research topics**

**Main Research Topics**

|                                        |                        |
|----------------------------------------|------------------------|
| 2023 Global Equity Derivatives Outlook | <a href="#">07-Dec</a> |
| Volatility Knock-Out Put Options       | <a href="#">08-Feb</a> |
| Latest Structured Product Update       | <a href="#">01-Feb</a> |

**Main Macro Trade Ideas**

|                                                                   |                     |                        |
|-------------------------------------------------------------------|---------------------|------------------------|
| Buy DAX Jun-23 put spread collar                                  |                     | <a href="#">05-Apr</a> |
| Buy DEDZ4 130 140 call spread vs. 115 put                         |                     | <a href="#">29-Mar</a> |
| SX7E Apr-23 call flies and call ratios                            |                     | <a href="#">15-Mar</a> |
| Sell VSTOXX March 17 puts                                         |                     | <a href="#">08-Mar</a> |
| Buy VSTOXX April 21-23 call spreads and sell VSTOXX March 17 puts |                     | <a href="#">08-Mar</a> |
| SX7E 2024 div futures                                             |                     | <a href="#">01-Mar</a> |
| SX8P and SX7E 1Y dispersion                                       |                     | <a href="#">15-Feb</a> |
| Dual Digital SX5E up, EURUSD up                                   |                     | <a href="#">08-Feb</a> |
| Euro STOXX 50 Apr-23 call spreads                                 |                     | <a href="#">08-Feb</a> |
| Worst-of call NDX/SX5E/KOSPI2                                     |                     | <a href="#">08-Feb</a> |
| Euro STOXX 50 Jun-23 VKO puts                                     |                     | <a href="#">08-Feb</a> |
| SX5E Jun-23 107% call vs 90% put, knock-in at 65%                 |                     | <a href="#">04-May</a> |
| SX5E Dec23-Dec-24 FVA                                             |                     | <a href="#">16-Feb</a> |
| SX7E 2023 div futures                                             | Initiated 15-Sep-21 | <a href="#">07-Dec</a> |
| SX5E 2023 div futures                                             | Initiated 7-Jun-21  | <a href="#">07-Dec</a> |

Source: J.P. Morgan Quantitative and Derivatives Strategy

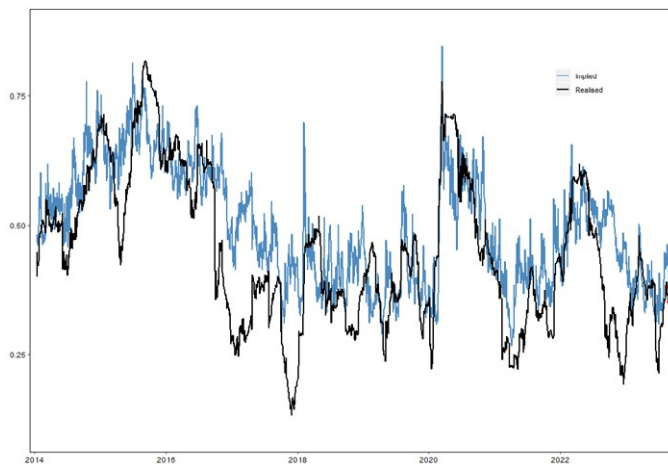
## Overview on dispersion

### Evolution of the Euro STOXX 50 correlation and comparison to the S&P 500

Dispersion has become a popular strategy with decent delivery of PnL in recent years. We examine the evolution of the index volatility to highlight the regime change that contributes towards this since 2018 and 2022.

**Figure 1: Implied and Realised Correlation 3M**

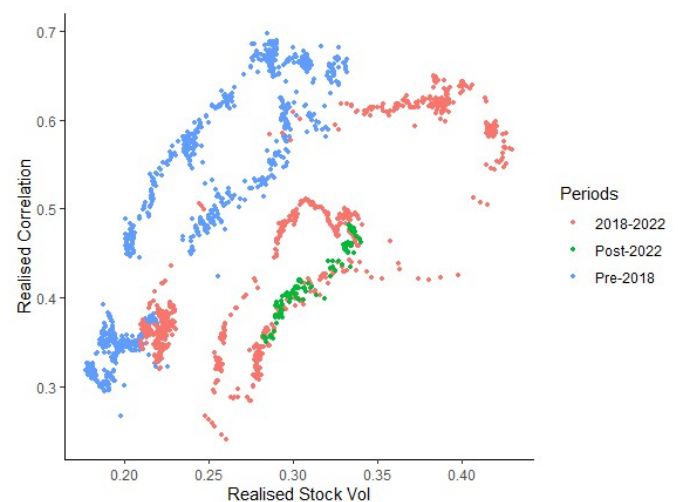
10Y history SX5E real-time constituents



Source: J.P. Morgan Equity Derivatives Strategy

**Figure 2: Realised Correlation versus Realised Stock Vol 12M**

10Y history SX5E real-time constituents

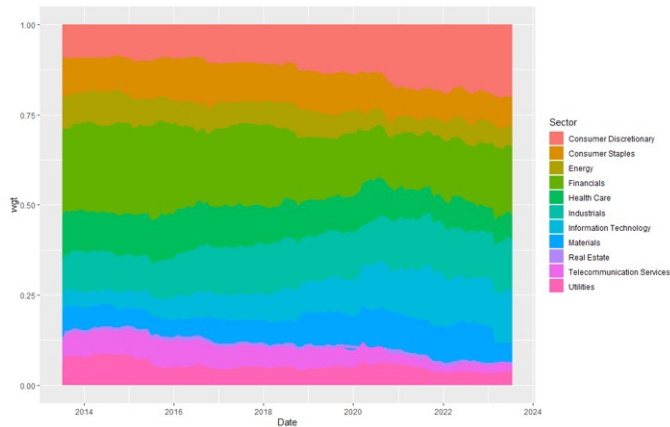


Source: J.P. Morgan Equity Derivatives Strategy

Over the long-term, the Euro STOXX 50 correlation has been on a declining trend, interrupted by spikes driven by market corrections such as the 2020 COVID and the 2022 Ukraine war crashes. This trend is consistent with the long-term evolution of **sector weights (Figure 3)** and **style factors (Figure 4)**.

The Euro STOXX 50 became a more diversified index as it reduced the heavy weight in banks from 2018 onwards, as discretionary names (e.g. luxury stocks) and Technology increased in weight. Contextually, style factor loading became less concentrated, as Quality and Growth became less negative at an index level and as the weight of Value decreased.

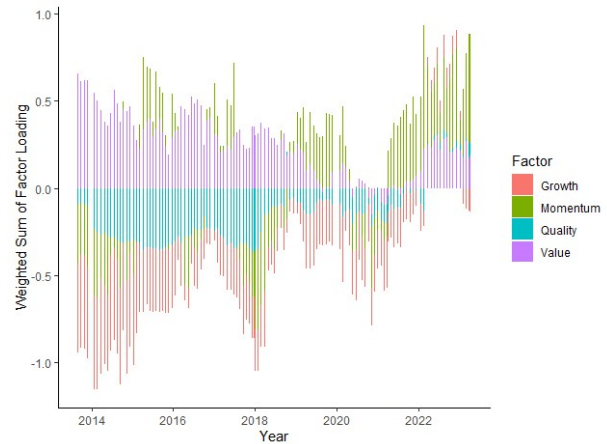
Figure 3: Euro STOXX 50 index sector split



Source: J.P. Morgan Equity Derivatives Strategy

Figure 4: Euro STOXX 50 index factor loading

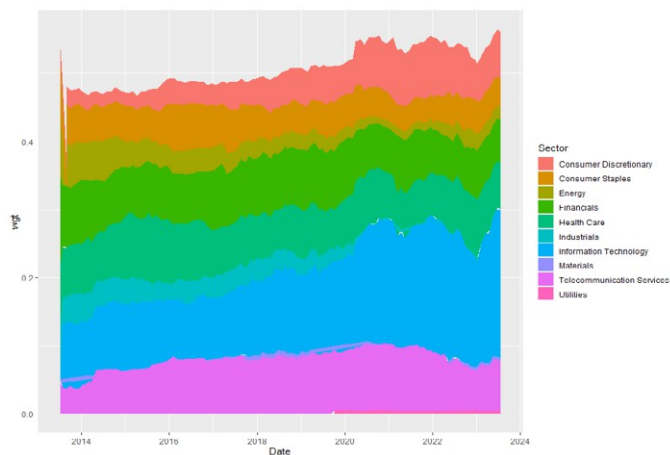
Normalised by MSCI ACWI



Source: J.P. Morgan Equity Derivatives Strategy

In contrast, the S&P 500 index became less diversified across sectors due to the strong out-performance of Tech, which makes-up almost half of the weight in the top 50 names in the S&P 500 (IT and Telecom sector). This change also led to an increasingly larger Quality tilt and progressively more negative Value exposure, which increased the factor sensitivity overall as Value and Quality are usually negatively correlated.

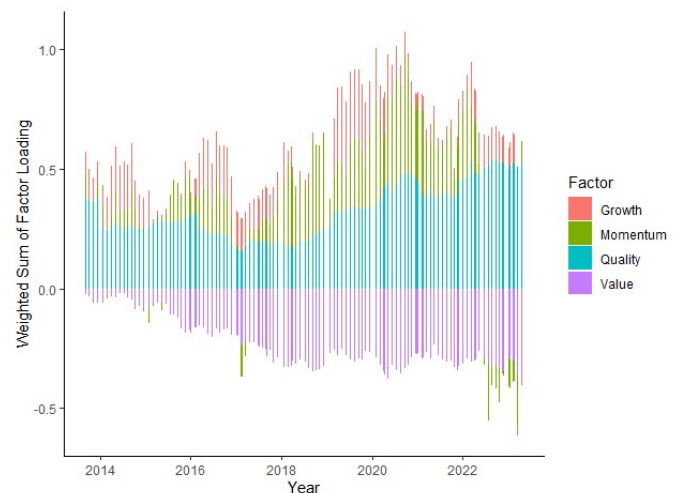
Figure 5: SPX Top 50 index weights by sector



Source: J.P. Morgan Equity Derivatives Strategy

Figure 6: SPX index factor loading

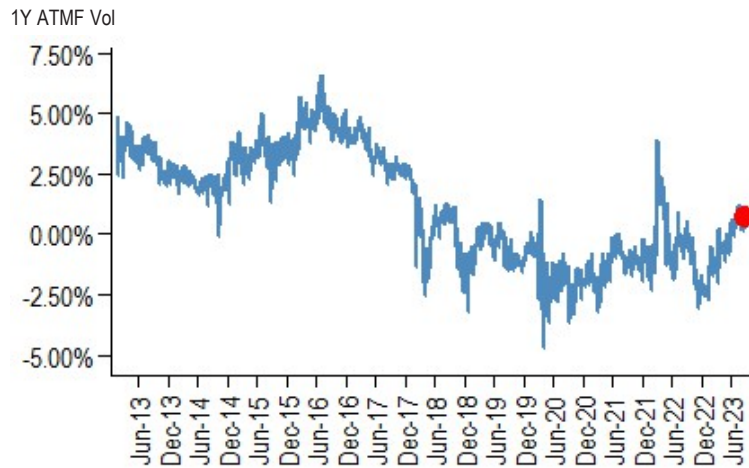
Top100 names only, Normalised by MSCI ACWI



Source: J.P. Morgan Equity Derivatives Strategy

Another indication of this dynamic is the fact that the implied volatility on Euro STOXX 50 index started **trading at a discount** versus S&P since 2018 (ex 2020 and 2022 market crashes) as the realised levels trended lower.

Figure 7: Euro STOXX 50 less S&P 500 implied volatility spread

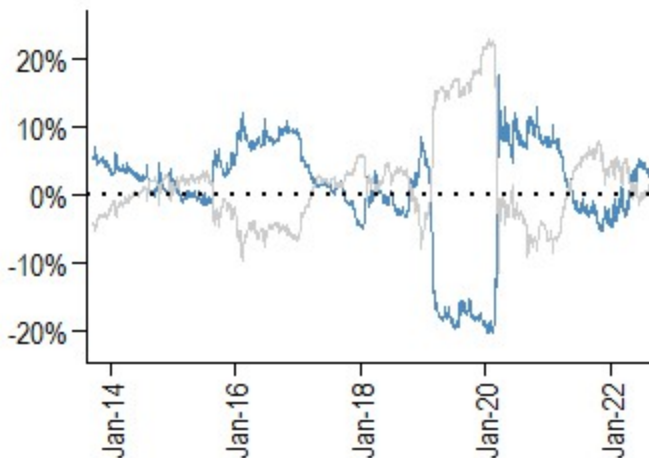


Source: J.P. Morgan Equity Derivatives Strategy

The two charts below (Figure 8 and Figure 9) show the index and single-stock PnL contributions to Euro STOXX 50 and S&P 500 dispersion trades separately. For trades entered after 2022, **the Euro STOXX 50 delivered more PnL on the index leg versus S&P, while for the stock leg the S&P names delivered more versus Euro STOXX names.**

Figure 8: S&P 500 top 50 dispersion PnL split between index and single-stock leg

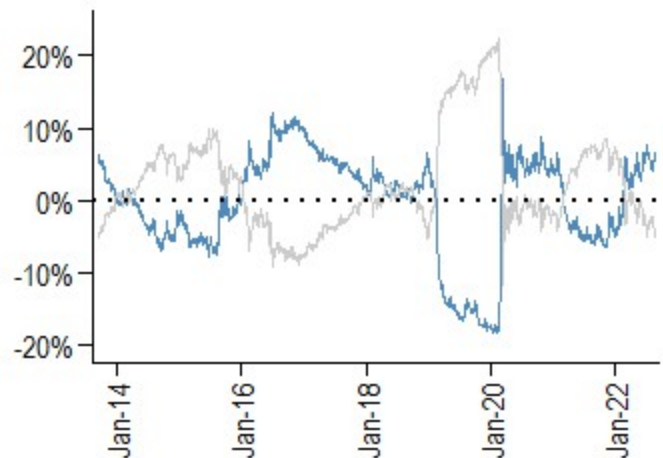
Assuming entry level at 1Y ATM vol, 1Y Tenor, Plot against Trade Date  
 Blue Line Index PnL, Grey Line Stock PnL



Source: J.P. Morgan Equity Derivatives Strategy

Figure 9: Euro STOXX 50 top 47 dispersion PnL split between index and single-stock leg

Assuming entry level at 1Y ATM vol, 1Y Tenor, Plot against Trade Date  
 Blue Line Index PnL, Grey Line Stock PnL



Source: J.P. Morgan Equity Derivatives Strategy

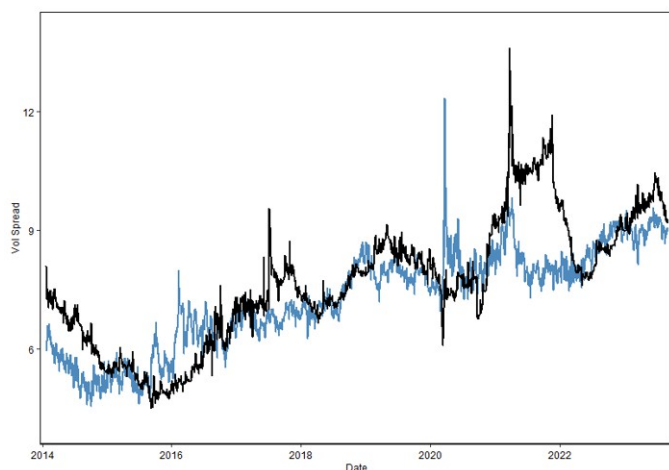
### Three challenges for European dispersion trading

Harvesting European dispersion has become more challenging after a positive streak of several years of solid returns. Anecdotally, this has happened following the launch of several new dispersion pods or funds late last year. We highlight **three factors which may have led to this challenge** and in the rest of the report recommend potential methods of improvements.

The first challenge has been the **entry level** - the decent delivery of dispersion and crowding in the space has kept the index-single stock volatility spread elevated (Figure 10). In addition to this, the divergence between index and single stock implied skew from 2022 made the downside vol spread even wider (Figure 11).

**Figure 10: SX5E (ex-6) 1Y dispersion vol spread**

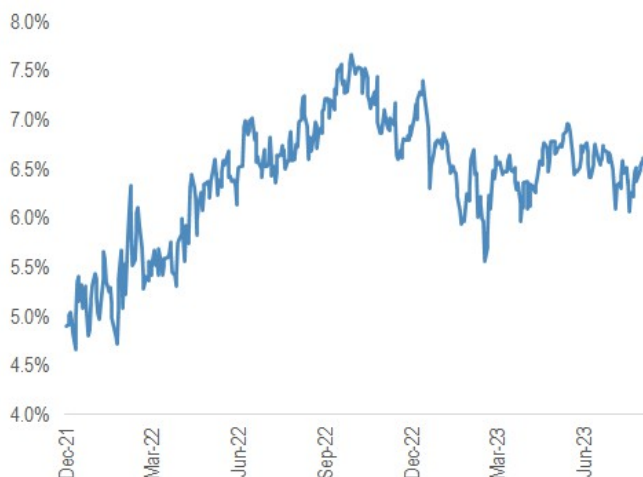
Blue Line Implied Vol Spread (Assume ATMV+1.2%), Black Line Realised Vol Spread



Source: J.P. Morgan Equity Derivatives Strategy

**Figure 11: Euro STOXX 50 90% strike stock-index implied volatility spread**

1Y Tenor, SX5E Ex-6 names

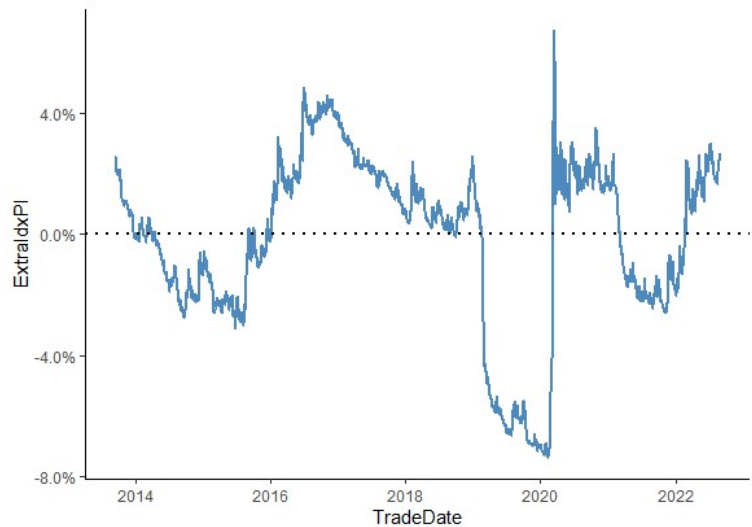


Source: J.P. Morgan Equity Derivatives Strategy

The second challenge has been the **vega-notional weighting scheme**. As most dispersion traders use a vega neutral position, the amount of correlation sold was not enough to capitalize on the large drop in realized correlation since 2022. Figure 12 shows the PnL for selling a [1.4x index vega notional](#); the Euro STOXX 50 dispersion PnL would have improved by 2 vol points.

**Figure 12: Excess PnL for Euro STOXX 50 dispersion when selling 1.4x index notional vs. 1x**

1Y tenor, Realtime constituents, ex-6 names, versus Trade Start Date

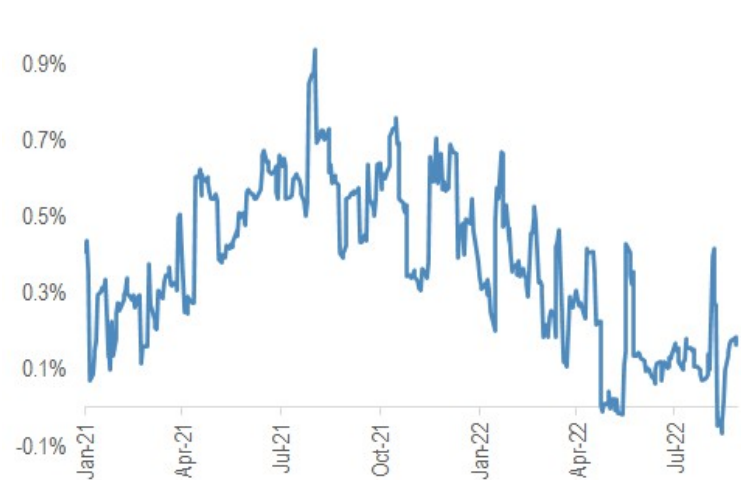


Source: J.P. Morgan Equity Derivatives Strategy

The last challenge was the impact of the **6 commonly excluded names** from Euro STOXX 50 dispersion. Despite being small in weight, these stocks delivered much more volatility versus the re-weighted 44 names basket. The contribution for trades launched since 2021 has averaged to around 0.4 vol points and was 0.22 vol points after 2022.

**Figure 13: Excess volatility for full Euro STOXX 50 vs. re-weighted basket with 6 exclusions**

1Y tenor, Vol PnL versus Trade Start Date



Source: J.P. Morgan Equity Derivatives Strategy

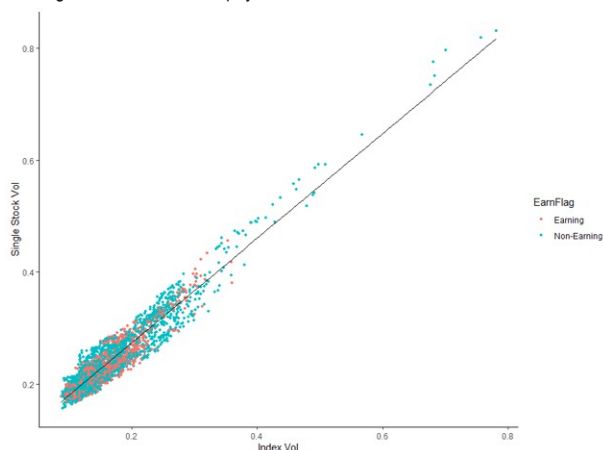
## Improvements from Trading Earnings

### Index and Stock Volatility Ratio

On close examination of the [Stock and Index volatility ratio](#), we highlight the seasonality on dispersion during earnings period. The 1M tenor showed a clearer picture on the spread. The implied beta showed 1.39x for Earnings months versus 1.35x, while the realized beta was 1.45x versus 1.3x. In summary, **while the market is indeed pricing in an “earnings premium” in the relationship between single stock and index implied volatility, this effect is underpriced relative to realized.**

**Figure 14: Single stocks vs index 1M implied volatility over earnings and non-earnings periods**

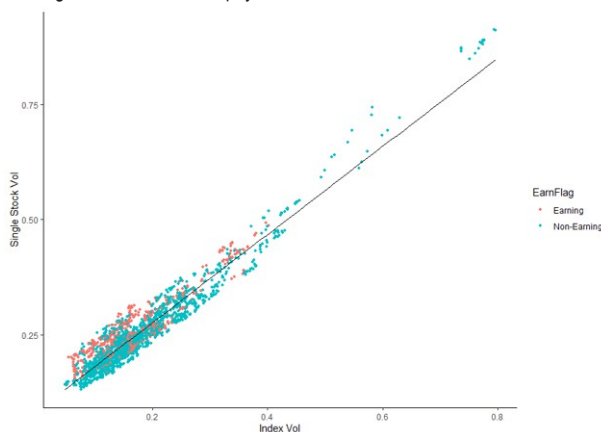
Earnings Periods Include Expiry Dates in 2,5,8,11 Months of the Year



Source: J.P. Morgan Equity Derivatives Strategy

**Figure 15: Single stocks vs index 1M realized volatility over earnings and non-earnings periods**

Earnings Periods Include Expiry Dates in 2,5,8,11 Months of the Year

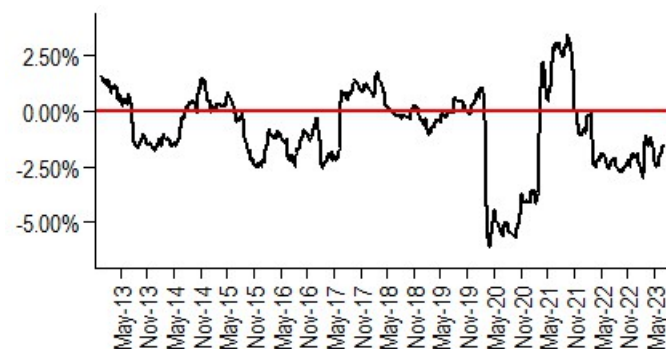


Source: J.P. Morgan Equity Derivatives Strategy

We see the effect of higher dispersion during earnings for long-dated realized volatilities too. **Index volatility realizes lower** if we include earnings periods and certain single-stock volatilities realize higher.

**Figure 16: Euro STOXX 50 Index realized volatility spread of normal vol minus annualized vol ex earning periods**

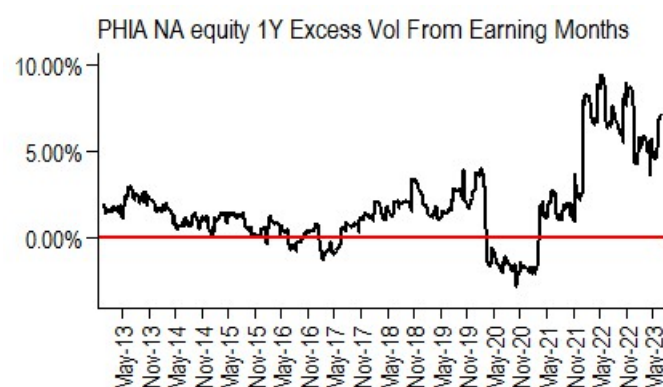
1Y rolling, earning period is 10th Jan/Apr/Jul/Oct for 5.5 weeks each



Source: J.P. Morgan Equity Derivatives Strategy

**Figure 17: Philips realized volatility spread of normal vol minus annualized vol ex earning periods**

1Y rolling, earning period is 10th Jan/Apr/Jul/Oct for 5.5 weeks each



Source: J.P. Morgan Equity Derivatives Strategy

On the index side, we recommend **selling more index leg during earnings period subject to seasonality** (discussed later), and we recommend potentially overlaying VSTOXX zero-cost call spread collar to hedge against volatility spikes.

**Figure 18: Selling More Index Vega notional (1.4x) over earnings periods leads to a sharp improvement in median PnL**

Compare with sell 1.4x index during earnings, result Ex 2020, 2022 to avoid distortion, Assume PnL is realized minus 1M implied ATM spread

|               | Vega Neutral | 1.4x Earning | 1.4x Always |
|---------------|--------------|--------------|-------------|
| <b>Median</b> | 0.28%        | 0.49%        | 0.57%       |
| <b>95VaR</b>  | -2.61%       | -4.28%       | -5.80%      |

Source: J.P. Morgan Quantitative and Derivatives Strategy

We can see the extra 21 bps pick up versus the vega neutral position if we sold 1.4x index notional **during earnings only**. If we sold 1.4x index leg all the time, there is only an additional 8 bps pick up, but the 95VaR is also much worse.

## Earnings Sensitive Stocks and One Day Jump Effect

On the stock side we highlight the importance to own gamma on more [“earnings sensitive stocks” during the reporting season](#). Figure 20 and Figure 20 show large cap EU and US stocks which historically displayed a significant earnings volatility premium over background realized volatility over a 10-year backtest period.

**Figure 19: US names with excess vol in earnings period**

Hit ratio (excess vol positive) is 1Y rolling, Ref vol is average realised vol

| Ticker   | Excess Vol | Hit Ratio | Ref Vol | Sector                 |
|----------|------------|-----------|---------|------------------------|
| AMZN US  | 4.9%       | 92%       | 32%     | Consumer Discretionary |
| NFLX US  | 8.8%       | 91%       | 47%     | Communication Services |
| GOOGL US | 2.6%       | 90%       | 26%     | Communication Services |
| IBM US   | 2.4%       | 90%       | 22%     | Information Technology |
| AMD US   | 6.2%       | 89%       | 57%     | Information Technology |
| EL US    | 1.5%       | 86%       | 26%     | Consumer Staples       |
| EA US    | 3.1%       | 84%       | 31%     | Communication Services |
| MSFT US  | 1.3%       | 83%       | 26%     | Information Technology |
| UPS US   | 2.0%       | 83%       | 22%     | Industrials            |
| GILD US  | 1.9%       | 83%       | 27%     | Health Care            |
| META US  | 4.7%       | 83%       | 37%     | Communication Services |
| PYPL US  | 3.0%       | 82%       | 37%     | Financials             |
| V US     | 0.9%       | 81%       | 24%     | Financials             |
| AAPL US  | 1.1%       | 80%       | 28%     | Information Technology |
| SBUX US  | 0.9%       | 80%       | 25%     | Consumer Discretionary |
| INTC US  | 1.5%       | 79%       | 29%     | Information Technology |
| DIS US   | 1.0%       | 79%       | 24%     | Communication Services |
| CAT US   | 0.9%       | 77%       | 28%     | Industrials            |
| NVDA US  | 1.5%       | 76%       | 42%     | Information Technology |
| LRCX US  | 1.0%       | 76%       | 37%     | Information Technology |
| ABBV US  | 1.2%       | 76%       | 27%     | Health Care            |
| LMT US   | 0.5%       | 75%       | 21%     | Industrials            |
| NOC US   | 0.6%       | 75%       | 22%     | Industrials            |
| TMO US   | 0.6%       | 73%       | 23%     | Health Care            |
| FCX US   | 1.3%       | 73%       | 50%     | Materials              |
| MRK US   | 1.0%       | 73%       | 21%     | Health Care            |
| NEM US   | 1.0%       | 72%       | 36%     | Materials              |
| NXPI US  | 1.4%       | 71%       | 38%     | Information Technology |

Source: J.P. Morgan Equity Derivatives Strategy

**Figure 20: EU names with excess vol in earnings period**

Hit ratio (excess vol positive) is 1Y rolling, Ref vol is average realised vol

| Ticker   | Excess Vol | Hit Ratio | Ref Vol | Sector                 |
|----------|------------|-----------|---------|------------------------|
| LOGN SW  | 5.3%       | 91%       | 33%     | Information Technology |
| NOKIA FH | 4.2%       | 88%       | 37%     | Information Technology |
| LONN SW  | 1.9%       | 86%       | 27%     | Health Care            |
| GIVN SW  | 1.2%       | 86%       | 19%     | Materials              |
| VWS DC   | 2.8%       | 84%       | 43%     | Industrials            |
| PHIA NA  | 1.8%       | 83%       | 26%     | Health Care            |
| SAN FP   | 0.9%       | 82%       | 22%     | Health Care            |
| ENR GR   | 2.1%       | 82%       | 39%     | Industrials            |
| DSY FP   | 1.6%       | 82%       | 24%     | Information Technology |
| SAP GR   | 1.9%       | 82%       | 24%     | Information Technology |
| ASML NA  | 1.3%       | 81%       | 30%     | Information Technology |
| GEBN SW  | 1.7%       | 80%       | 21%     | Industrials            |
| ABBN SW  | 1.1%       | 80%       | 22%     | Industrials            |
| SCHP SW  | 1.3%       | 78%       | 20%     | Industrials            |
| ADS GR   | 1.5%       | 76%       | 28%     | Consumer Discretionary |
| STMPA FP | 1.9%       | 75%       | 38%     | Information Technology |
| HEN3 GR  | 0.8%       | 73%       | 21%     | Consumer Staples       |
| CAP FP   | 0.4%       | 72%       | 28%     | Information Technology |
| AZN LN   | 1.7%       | 72%       | 23%     | Health Care            |
| MTX GR   | 0.3%       | 71%       | 29%     | Industrials            |
| ABI BB   | 0.7%       | 71%       | 26%     | Consumer Staples       |
| SIE GR   | 0.2%       | 70%       | 24%     | Industrials            |
| AFX GR   | 0.4%       | 69%       | 29%     | Health Care            |
| GSK LN   | 1.0%       | 69%       | 19%     | Health Care            |
| MRK GR   | 0.4%       | 68%       | 24%     | Health Care            |
| UHR SW   | 1.3%       | 68%       | 27%     | Consumer Discretionary |
| BARN SW  | 0.7%       | 68%       | 19%     | Consumer Staples       |
| MC FP    | 0.6%       | 67%       | 26%     | Consumer Discretionary |

Source: J.P. Morgan Equity Derivatives Strategy

As most dispersion trades are positioned for longer maturity, we check the effect of a one-day jump on the 1-year tenor realized volatility and find a **significant pick-up in realized vol for lower vol stock**.

**Figure 21: Impact of one-day jump on 1-year realized volatility**

Assume realised vol at the "vol ref" levels on the left index for other days

| Vol ref \ Jump | 15%  | 10%  | 5%   | -5%  | -10% | -15% |
|----------------|------|------|------|------|------|------|
| 60.0%          | 1.5% | 0.6% | 0.1% | 0.1% | 0.8% | 2.0% |
| 50.0%          | 1.8% | 0.8% | 0.1% | 0.2% | 1.0% | 2.5% |
| 40.0%          | 2.3% | 1.0% | 0.2% | 0.2% | 1.3% | 3.1% |
| 30.0%          | 3.0% | 1.4% | 0.3% | 0.4% | 1.7% | 4.1% |
| 20.0%          | 4.4% | 2.1% | 0.5% | 0.6% | 2.6% | 5.7% |

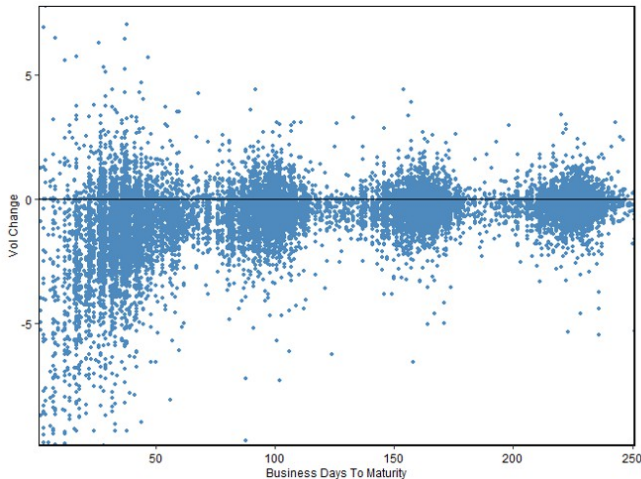
Source: J.P. Morgan Equity Derivatives Strategy

## Earnings Volatility Reset

The other side of the coin when owning gamma on earnings names, despite the risk premium being often underpriced, is that after the event there tends to be a reset in the implied volatility. We take a closer look at the effect of **post earnings volatility reset** per name and a practical scenario table for the **balance of PnL from gamma and vol reset**.

**Figure 22: Vol Reset by Option Tenor**

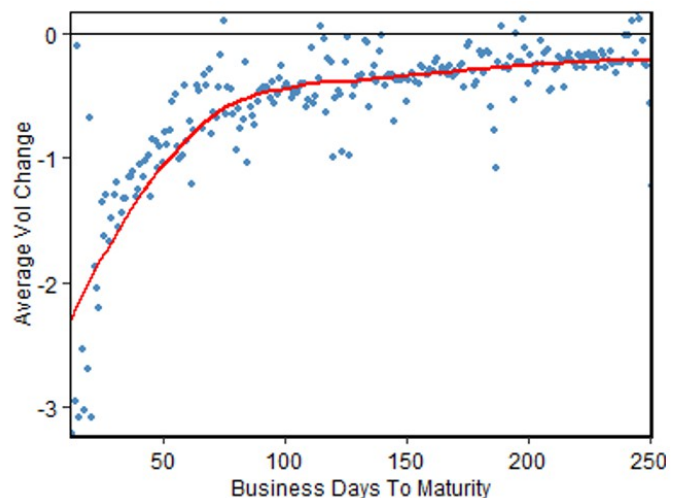
EU optional stocks, 10Y history fixed strike ATM Vol change on earnings day and post



Source: J.P. Morgan Equity Derivatives Strategy

**Figure 23: Average Vol Reset by Option Tenor**

EU optional stocks, 10Y history fixed strike ATM Vol change on earnings day and post



Source: J.P. Morgan Equity Derivatives Strategy

**Interestingly and importantly the vol reset is not limited to front month options but happens across the curve and it is subject to seasonality.** As US names generally have higher average implied volatility they are subject to larger re-sets. However, **the realized earnings day move to implied vol ratio is in fact higher in the US than Europe, making the US more desirable for those wishing to own earnings gamma.**

For both continents, stocks have the **largest earnings day moves during Q4 reporting (Jan)**. In Europe, Q2 (Jul) represents the best value of realized move/implied vol ratio. In the US, Q3 (Oct) provides better value to own gamma as it exhibits the largest earnings day spot move/implied vol period.

Figure 24: European top 130 option liquid stocks earnings move and vol reset seasonality

| Season    | Fixed Strike Vol Reset On Eday |        |        |        |        | Avg Vol Prior Eday |        |        |        |        |
|-----------|--------------------------------|--------|--------|--------|--------|--------------------|--------|--------|--------|--------|
|           | Abs Spot Move                  | 2M     | 5M     | 8M     | 11M    | Avg Vol            | 2M     | 5M     | 8M     | 11M    |
| Q1 (Apr)  | 2.50%                          | -1.15% | -0.40% | -0.24% | -0.18% | 25.80%             | 26.60% | 25.60% | 25.60% | 25.50% |
| Q2 (Jul)  | 2.90%                          | -1.24% | -0.45% | -0.31% | -0.22% | 25.00%             | 24.90% | 25.00% | 24.80% | 25.10% |
| Q3 (Oct)  | 2.70%                          | -1.05% | -0.47% | -0.29% | -0.18% | 25.80%             | 26.20% | 25.70% | 25.70% | 25.60% |
| Q4 (Jan)  | 3.00%                          | -1.53% | -0.54% | -0.34% | -0.22% | 26.70%             | 27.80% | 26.70% | 26.00% | 26.30% |
| Average   | 2.80%                          | -1.25% | -0.46% | -0.30% | -0.20% |                    | 26.40% | 25.80% | 25.50% | 25.60% |
| Hit Ratio |                                | 78%    | 72%    | 66%    | 68%    |                    |        |        |        |        |

Source: J.P. Morgan Equity Derivatives Strategy

Figure 25: US top 250 mkt cap stocks earnings move and vol reset seasonality

Showing Q1 ex-2020 as "base case" due to the large distortion on data for avg vol

| Season           | Fixed Strike Vol Reset On Eday |        |        |        |        | Avg Vol Prior Eday |        |        |        |        |
|------------------|--------------------------------|--------|--------|--------|--------|--------------------|--------|--------|--------|--------|
|                  | Abs Spot Move                  | 2M     | 5M     | 8M     | 11M    | Avg Vol            | 2M     | 5M     | 8M     | 11M    |
| Q1 (Apr) All     | 3.80%                          | -2.10% | -0.80% | -0.50% | -0.40% | 31.94%             | 32.79% | 31.89% | 31.63% | 31.44% |
| Q1 (Apr) Ex-2020 | 3.60%                          | -2.10% | -0.70% | -0.40% | -0.30% | 27.45%             | 27.22% | 27.28% | 27.58% | 27.72% |
| Q2 (Jul)         | 3.62%                          | -2.00% | -0.80% | -0.50% | -0.30% | 29.18%             | 29.01% | 29.15% | 29.27% | 29.30% |
| Q3 (Oct)         | 3.80%                          | -1.90% | -0.70% | -0.40% | -0.30% | 29.24%             | 29.53% | 29.19% | 29.12% | 29.12% |
| Q4 (Jan)         | 3.89%                          | -2.10% | -0.70% | -0.40% | -0.30% | 30.34%             | 31.35% | 30.22% | 29.91% | 29.89% |
| Average          | 3.73%                          | -2.03% | -0.73% | -0.44% | -0.32% |                    | 29.28% | 28.96% | 28.97% | 29.01% |
| Hit Ratio        |                                | 84%    | 74%    | 67%    | 64%    |                    |        |        |        |        |

Source: J.P. Morgan Equity Derivatives Strategy

We highlight the names with large volatility resets historically in the tables below.

Figure 26: US earnings day top vol reset names

5Y backtest till 2022

|       | Avg 2M Vol Pre | Avg Eday Move | 2M     | 5M     | 1Y     |
|-------|----------------|---------------|--------|--------|--------|
| NFLX  | 42.00%         | 8.40%         | -5.80% | -2.10% | -0.80% |
| AMD   | 53.30%         | 7.50%         | -5.70% | -1.90% | -1.00% |
| TSLA  | 58.20%         | 6.90%         | -4.50% | -1.70% | -1.00% |
| ORCL  | 29.70%         | 5.60%         | -4.40% | -1.80% | -0.90% |
| META  | 35.40%         | 7.30%         | -4.30% | -1.50% | -0.80% |
| NVDA  | 45.70%         | 5.00%         | -4.10% | -1.30% | -0.60% |
| DIS   | 28.80%         | 2.80%         | -3.90% | -1.60% | -1.00% |
| CSCO  | 26.70%         | 5.40%         | -3.70% | -1.40% | -0.80% |
| PYPL  | 36.30%         | 6.90%         | -3.30% | -1.50% | -0.80% |
| AMZN  | 32.50%         | 5.50%         | -3.30% | -1.20% | -0.50% |
| AMAT  | 38.80%         | 3.90%         | -3.30% | -1.40% | -0.80% |
| GOOGL | 27.90%         | 4.70%         | -3.20% | -1.20% | -0.50% |
| CRM   | 33.70%         | 5.00%         | -3.10% | -1.30% | -0.80% |
| IBM   | 23.90%         | 5.40%         | -3.00% | -1.00% | -0.30% |
| DE    | 30.80%         | 4.30%         | -3.00% | -1.20% | -0.80% |
| EL    | 29.10%         | 5.40%         | -2.90% | -0.80% | -0.10% |
| LOW   | 29.70%         | 4.60%         | -2.80% | -1.00% | -0.40% |
| AVGO  | 35.20%         | 3.60%         | -2.80% | -1.10% | -0.60% |
| TGT   | 31.30%         | 7.80%         | -2.80% | -0.70% | -0.10% |
| TJX   | 29.80%         | 3.90%         | -2.70% | -0.70% | -0.20% |
| GE    | 38.70%         | 4.90%         | -2.70% | -1.00% | -0.50% |
| QCOM  | 34.50%         | 5.60%         | -2.70% | -0.80% | -0.50% |
| INTC  | 30.00%         | 6.70%         | -2.60% | -0.80% | -0.60% |
| TXN   | 28.20%         | 4.20%         | -2.60% | -1.10% | -0.70% |
| NOW   | 39.60%         | 5.50%         | -2.50% | -0.60% | 0.00%  |

Source: J.P. Morgan Equity Derivatives Strategy

Figure 27: EU earnings day top vol reset names

5Y backtest till 2022

|          | Avg 2M Vol Pre | Avg Eday Move | 2M     | 5M     | 1Y     |
|----------|----------------|---------------|--------|--------|--------|
| ADS GY   | 28.00%         | 5.50%         | -4.40% | -1.20% | -0.70% |
| NOKIA FH | 35.00%         | 7.60%         | -3.20% | -1.40% | -0.70% |
| BAER SW  | 26.00%         | 3.00%         | -2.60% | -1.20% | -0.80% |
| ACA FP   | 27.00%         | 2.10%         | -2.60% | -1.00% | -0.60% |
| STM FP   | 37.00%         | 5.00%         | -2.50% | -1.00% | -0.50% |
| MT NA    | 40.00%         | 3.50%         | -2.20% | -0.90% | -0.50% |
| IFX GY   | 33.00%         | 3.50%         | -2.20% | -0.90% | -0.50% |
| PHIA NA  | 24.00%         | 4.00%         | -2.00% | -0.70% | -0.40% |
| DSM NA   | 22.00%         | 2.50%         | -2.00% | -1.10% | -0.60% |
| AIR FP   | 32.00%         | 2.50%         | -2.00% | -0.60% | -0.30% |
| VIV FP   | 27.00%         | 2.80%         | -1.90% | -1.00% | -0.40% |
| ASML NA  | 28.00%         | 3.40%         | -1.70% | -0.80% | -0.50% |
| LXS GY   | 30.00%         | 3.70%         | -1.70% | -0.60% | -0.40% |
| 1COV GY  | 30.00%         | 2.20%         | -1.60% | -0.60% | -0.60% |
| BAYN GY  | 28.00%         | 2.90%         | -1.60% | -0.50% | -0.40% |
| ISP IM   | 26.00%         | 1.80%         | -1.50% | -0.10% | -0.20% |
| SOLB BB  | 25.00%         | 3.60%         | -1.50% | -0.80% | -0.40% |
| AZN LN   | 25.00%         | 3.80%         | -1.40% | -0.80% | -0.50% |
| VOD LN   | 25.00%         | 4.30%         | -1.40% | -1.00% | -0.40% |
| BARC LN  | 30.00%         | 3.00%         | -1.40% | -0.70% | -0.60% |
| ABI BB   | 26.00%         | 4.50%         | -1.40% | -0.70% | -0.40% |
| CSGN SW  | 27.00%         | 3.00%         | -1.40% | -0.30% | -0.20% |
| EVK GY   | 23.00%         | 2.40%         | -1.40% | -0.50% | -0.40% |
| CBK GY   | 37.00%         | 3.40%         | -1.30% | -0.30% | -0.20% |
| TEF SM   | 24.00%         | 2.10%         | -1.30% | -0.50% | -0.20% |

Source: J.P. Morgan Equity Derivatives Strategy

To get a better picture of how the combination of vol reset and spot move result in PnL on earnings day, we construct a scenario table for 2 sets of assumptions - on the level of vol reset at historical average (1.25 for 2M and 0.46 for 5M as Figure 28), and if vol

reset is twice as large (Figure 29). For lower vol names, shorter maturity (e.g. 2M) is preferred due to the low cost of carry. For higher vol names, longer maturity (e.g. 5M) provides more cushion. A smoother pnl profile can be achieved with long gamma short vega (e.g. Long 2M short 5M).

Figure 28: Total Pnl scenario for Vol Reset=1.25 in 2M and 0.46 in 5M

| 2M expiry, Vol reset ref =1.25% |        |        |        |       |       |  |
|---------------------------------|--------|--------|--------|-------|-------|--|
| Vol \ Spot move                 | 0.00%  | 2.00%  | 4.00%  | 6.00% | 8.00% |  |
| 22.00%                          | -0.48% | -0.30% | 0.26%  | 1.18% | 2.43% |  |
| 24.00%                          | -0.49% | -0.32% | 0.19%  | 1.04% | 2.19% |  |
| 26.00%                          | -0.50% | -0.34% | 0.13%  | 0.91% | 1.99% |  |
| 28.00%                          | -0.51% | -0.36% | 0.08%  | 0.80% | 1.81% |  |
| 30.00%                          | -0.51% | -0.38% | 0.03%  | 0.71% | 1.65% |  |
| 32.00%                          | -0.52% | -0.40% | -0.02% | 0.62% | 1.51% |  |
| 34.00%                          | -0.53% | -0.41% | -0.05% | 0.55% | 1.39% |  |
| 36.00%                          | -0.54% | -0.43% | -0.09% | 0.48% | 1.27% |  |
| 38.00%                          | -0.54% | -0.44% | -0.12% | 0.42% | 1.17% |  |
| 5M expiry, Vol reset ref =0.46% |        |        |        |       |       |  |
| Vol \ Spot move                 | 0.00%  | 2.00%  | 4.00%  | 6.00% | 8.00% |  |
| 22.00%                          | -0.28% | -0.17% | 0.18%  | 0.77% | 1.60% |  |
| 24.00%                          | -0.29% | -0.18% | 0.14%  | 0.68% | 1.44% |  |
| 26.00%                          | -0.29% | -0.20% | 0.10%  | 0.60% | 1.30% |  |
| 28.00%                          | -0.30% | -0.21% | 0.07%  | 0.53% | 1.18% |  |
| 30.00%                          | -0.30% | -0.22% | 0.04%  | 0.47% | 1.08% |  |
| 32.00%                          | -0.31% | -0.23% | 0.01%  | 0.42% | 0.99% |  |
| 34.00%                          | -0.31% | -0.24% | -0.01% | 0.37% | 0.91% |  |
| 36.00%                          | -0.32% | -0.25% | -0.04% | 0.32% | 0.83% |  |
| 38.00%                          | -0.32% | -0.26% | -0.06% | 0.29% | 0.77% |  |
| Long 2M short 5M                |        |        |        |       |       |  |
| Vol \ Spot move                 | 0.00%  | 2.00%  | 4.00%  | 6.00% | 8.00% |  |
| 22.00%                          | -0.20% | -0.13% | 0.08%  | 0.41% | 0.83% |  |
| 24.00%                          | -0.20% | -0.14% | 0.05%  | 0.36% | 0.75% |  |
| 26.00%                          | -0.21% | -0.15% | 0.03%  | 0.31% | 0.69% |  |
| 28.00%                          | -0.21% | -0.15% | 0.01%  | 0.27% | 0.63% |  |
| 30.00%                          | -0.21% | -0.16% | -0.01% | 0.24% | 0.57% |  |
| 32.00%                          | -0.21% | -0.17% | -0.02% | 0.21% | 0.52% |  |
| 34.00%                          | -0.22% | -0.17% | -0.04% | 0.18% | 0.48% |  |
| 36.00%                          | -0.22% | -0.18% | -0.05% | 0.16% | 0.44% |  |
| 38.00%                          | -0.22% | -0.18% | -0.06% | 0.13% | 0.40% |  |

Source: J.P. Morgan Equity Derivatives Strategy

Figure 29: Total Pnl scenario for Vol Reset=2.5 in 2M and 0.92 in 5M

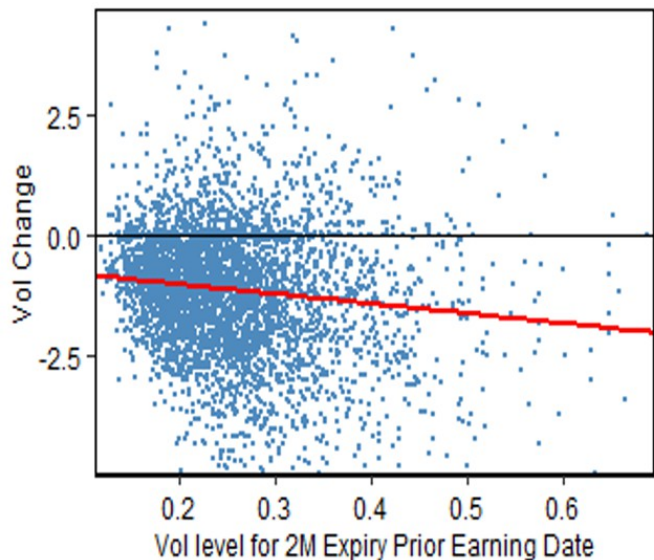
| 2M expiry, Vol reset ref =2.5%  |        |        |        |        |       |  |
|---------------------------------|--------|--------|--------|--------|-------|--|
| Vol \ Spot move                 | 0.00%  | 2.00%  | 4.00%  | 6.00%  | 8.00% |  |
| 22.00%                          | -0.89% | -0.70% | -0.11% | 0.86%  | 2.17% |  |
| 24.00%                          | -0.90% | -0.73% | -0.19% | 0.70%  | 1.91% |  |
| 26.00%                          | -0.91% | -0.75% | -0.26% | 0.57%  | 1.69% |  |
| 28.00%                          | -0.91% | -0.77% | -0.31% | 0.45%  | 1.49% |  |
| 30.00%                          | -0.92% | -0.79% | -0.37% | 0.34%  | 1.32% |  |
| 32.00%                          | -0.93% | -0.80% | -0.41% | 0.25%  | 1.17% |  |
| 34.00%                          | -0.94% | -0.82% | -0.45% | 0.17%  | 1.04% |  |
| 36.00%                          | -0.94% | -0.83% | -0.49% | 0.10%  | 0.92% |  |
| 38.00%                          | -0.95% | -0.85% | -0.52% | 0.03%  | 0.81% |  |
| 5M expiry, Vol reset ref =0.92% |        |        |        |        |       |  |
| Vol \ Spot move                 | 0.00%  | 2.00%  | 4.00%  | 6.00%  | 8.00% |  |
| 22.00%                          | -0.50% | -0.39% | -0.04% | 0.56%  | 1.40% |  |
| 24.00%                          | -0.51% | -0.41% | -0.08% | 0.47%  | 1.24% |  |
| 26.00%                          | -0.51% | -0.42% | -0.12% | 0.39%  | 1.10% |  |
| 28.00%                          | -0.52% | -0.43% | -0.15% | 0.31%  | 0.98% |  |
| 30.00%                          | -0.52% | -0.44% | -0.18% | 0.25%  | 0.87% |  |
| 32.00%                          | -0.53% | -0.45% | -0.21% | 0.20%  | 0.78% |  |
| 34.00%                          | -0.53% | -0.46% | -0.24% | 0.15%  | 0.69% |  |
| 36.00%                          | -0.54% | -0.47% | -0.26% | 0.10%  | 0.62% |  |
| 38.00%                          | -0.54% | -0.48% | -0.28% | 0.06%  | 0.55% |  |
| Long 2M short 5M                |        |        |        |        |       |  |
| Vol \ Spot move                 | 0.00%  | 2.00%  | 4.00%  | 6.00%  | 8.00% |  |
| 22.00%                          | -0.39% | -0.31% | -0.07% | 0.30%  | 0.76% |  |
| 24.00%                          | -0.39% | -0.32% | -0.11% | 0.24%  | 0.67% |  |
| 26.00%                          | -0.39% | -0.33% | -0.13% | 0.18%  | 0.59% |  |
| 28.00%                          | -0.40% | -0.34% | -0.16% | 0.13%  | 0.52% |  |
| 30.00%                          | -0.40% | -0.35% | -0.18% | 0.09%  | 0.45% |  |
| 32.00%                          | -0.40% | -0.35% | -0.20% | 0.05%  | 0.40% |  |
| 34.00%                          | -0.40% | -0.36% | -0.22% | 0.02%  | 0.34% |  |
| 36.00%                          | -0.41% | -0.36% | -0.23% | -0.01% | 0.30% |  |
| 38.00%                          | -0.41% | -0.37% | -0.24% | -0.03% | 0.26% |  |

Source: J.P. Morgan Equity Derivatives Strategy

We found higher vol stocks do not necessarily have larger resets (Figure 30) and for most stocks the implied vol bounces back by day 5 post the event (Figure 31). Therefore, it is worth **building long vega on stocks for “non-earning-sensitive names” on the day of earnings** to achieve an entry level likely to be relatively the lowest.

**Figure 30: Earnings day vol reset versus Vol of the stock**

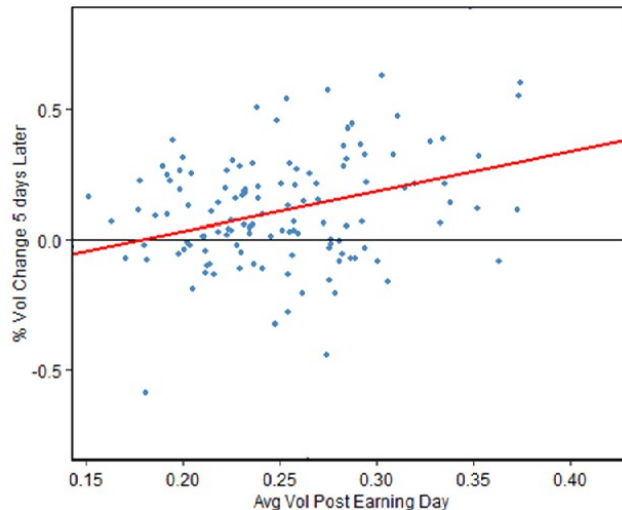
Fixed strike vol the day prior earnings for 2M tenor



Source: J.P. Morgan Equity Derivatives Strategy

**Figure 31: Vol bounces back 5 days post Earnings for longer tenor**

For all expiries greater than 2M, each point represent the avg change for one stock



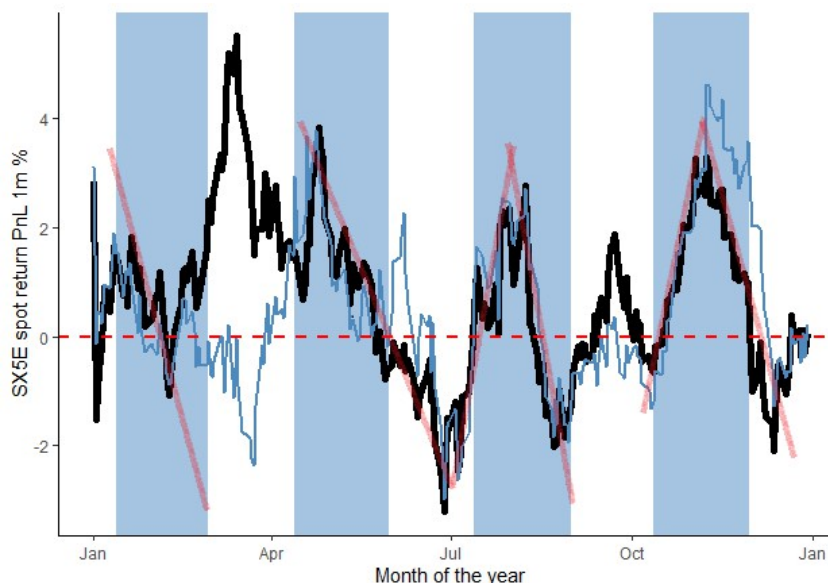
Source: J.P. Morgan Equity Derivatives Strategy

## Seasonality

Further improvement can be achieved if we are more tactical on the seasonality. We already saw above the seasonality on stock vol entry levels and reset by different earnings quarter. We highlight further findings below to improve tactical entry points.

**Figure 32: SX5E Index 1M Average Spot Return**

Blue line full 10Y data. Black line excludes 2020, 2022



Source: J.P. Morgan Equity Derivatives Strategy

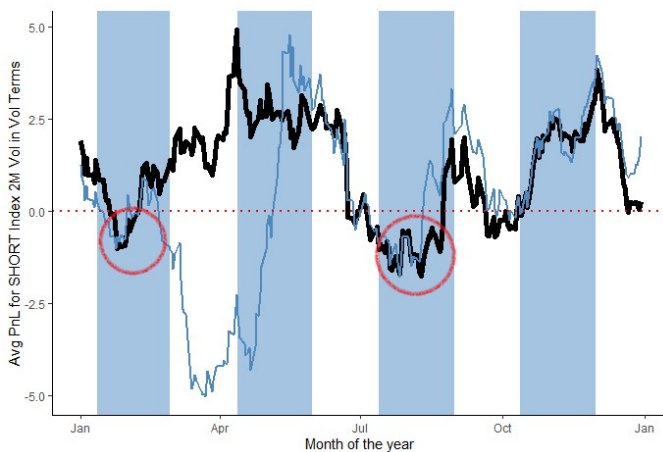
Seasonality of spot returns on SX5E showed that the market **tends to trade up into the earnings season, followed by a correction afterwards** (Figure 32). Looking at the split

between index and stock legs, we highlight the following findings:

1. Seasonality is clear for **Q4 (Jan)** and **Q2 (July)** where **short index vol produced negative PnL which was more than offset by higher single stock volatility**. This is coherent with our earlier finding that stocks historically deliver higher earnings day moves in those two quarters, but it indicates **implied volatility on stocks usually underprice earnings move** as vol PnL is positive even taking account of entry levels.
2. Q3 (Oct-Nov) dispersion PnL is largely **a function of the index leg**. The “**end-of-year seasonality**” could be a driver of the additional profit such as what we saw in 2022.
3. The Q1 (Apr) season rally in “normal markets” (black line) is coupled with short index vol working well and long stock vol failing to realize.

**Figure 33: Short SX5E Index Vol 2M PnL (Implied minus Realised)**

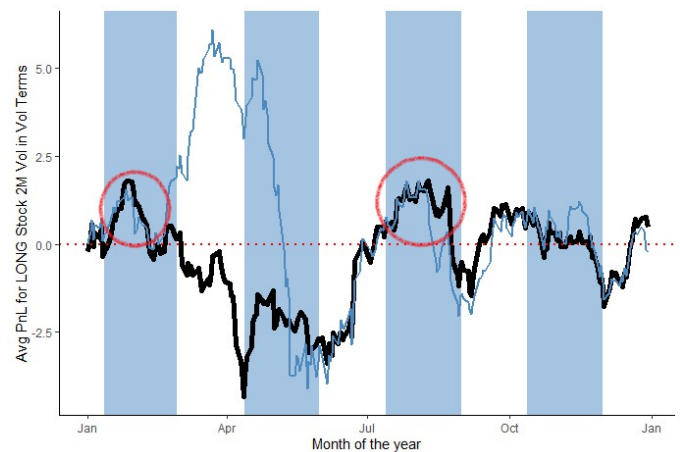
Blue line full 10Y data. Black line excludes 2020, 2022



Source: J.P. Morgan Equity Derivatives Strategy

**Figure 34: Long SX5E Component Stock Vol 2M PnL (Realised minus Implied)**

Blue line full 10Y data. Black line excludes 2020, 2022. Stocks leg index weighted

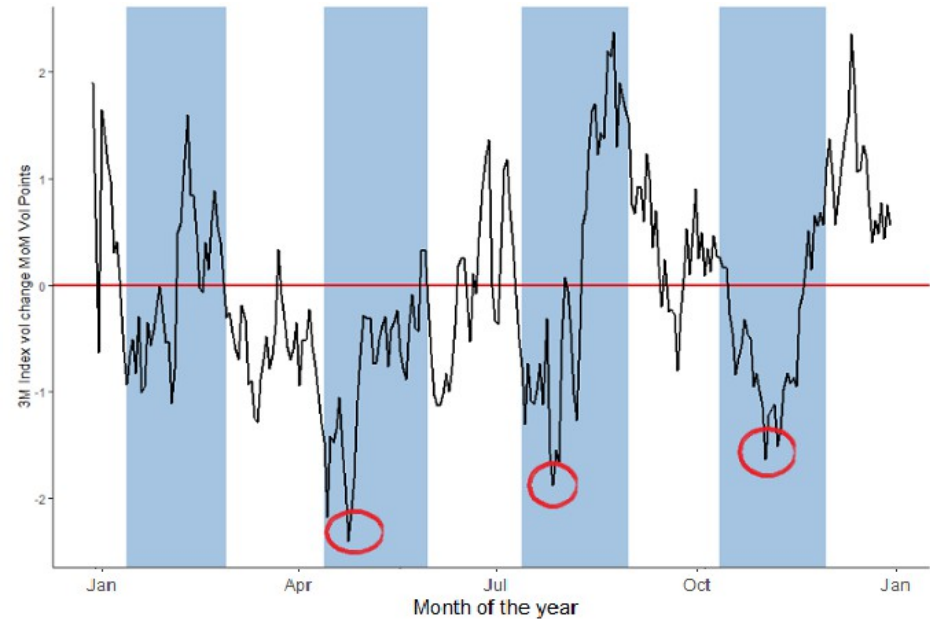


Source: J.P. Morgan Equity Derivatives Strategy

We note that, consistent with the spot moves (Figure 32), the seasonality on index implied volatility also **trades down into earnings and trough around 1 week post earning start** (Figure 35), and **revert by about one month later**. Although skew contributes to some of the moves (spot up into earnings), it describes the market behaviour and is very significant for Q1(Apr),Q2(Jul) and Q3(Oct).

Figure 35: 3M ATMF Vol on SX5E Month on Month Change

10Y data exclude 2020, 2022



Source: J.P. Morgan Equity Derivatives Strategy

## Other Potential Techniques to Improve PnL

We already discussed above some **techniques to improve the dispersion PnL** by putting on **tactical positions around earnings period**:

- Own gamma on **earnings sensitive names**, can structure as short dated calendar to minimize vol reset impact
- Build vega on earnings day **post vol reset**, especially for non earnings sensitive names
- **Short more index leg** into earnings, can start executing around 2-3 weeks before earnings start
- Trade along with **seasonality** to have bigger dispersion size for Q2 (July) and Q4 (Jan), more index short in late Q3.

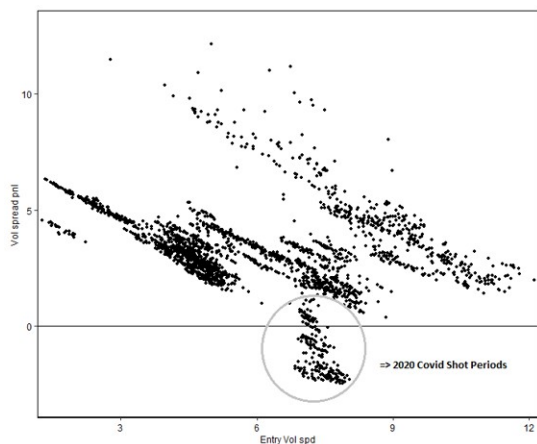
There are other potential technique to enhance PnL from achieving **better entry levels** and getting **better carry** through the life of the trade.

### Entry Level

We highlighted previously the issue with challenging entry level. The entry levels versus terminal payoff for the S&P 500 and Euro STOXX 50 showed a clear **inverse relationship** of PnL vs entry level volatility spread.

Figure 36: SPX Top 50 1Y tenor

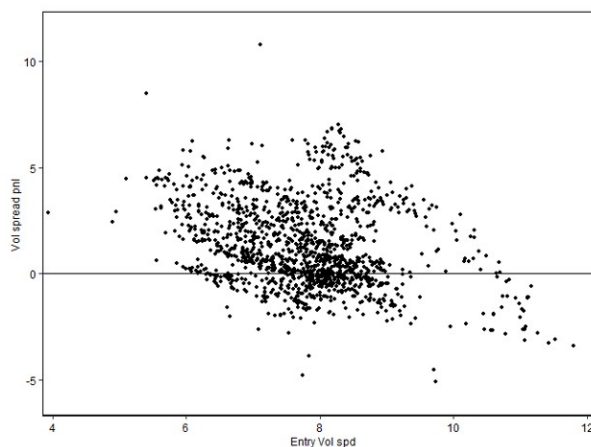
10Y backtest on realtime top50 names, entry vol spread is 1Y ATMF vol



Source: J.P. Morgan Quantitative and Derivatives Strategy

Figure 37: SX5E 50 3M tenor

5Y backtest on realtime full index, entry vol spread is 3M ATMF vol



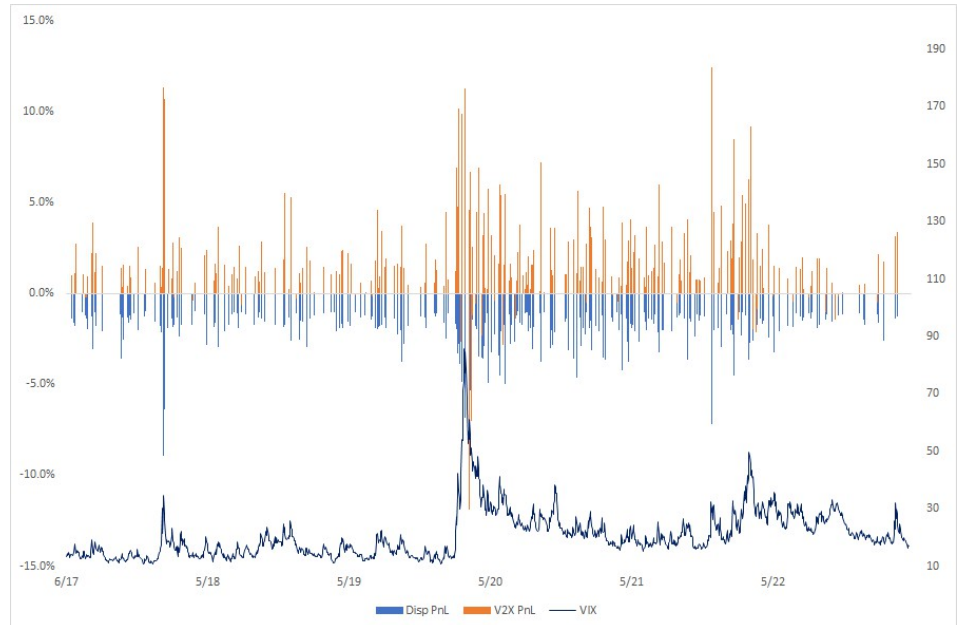
Source: J.P. Morgan Quantitative and Derivatives Strategy

We can improve the entry level by legging in some stock-index pairs as discussed earlier during earnings season. For new trades, one technique worth exploring when market condition suits is to overlay with VSTOXX call spread collars, especially when the dispersion book has more notional on the short index leg.

We highlight the “mirror image” between dispersion entry level (1Y ATMF implied vol spread as a proxy) and VSTOXX spot (Figure 38) , with a hedge ratio of 1x4 (1x V2X vega notional for 4x dispersion vega notional). We can also think of the entry level as the unrealised part of the dispersion PnL for 1Y tenor, the hit ratio we can have if to long a V2X spot “hedge” is 87%.

**Figure 38: Dispersion Unrealised PnL versus V2X “hedge” with 1x4 Ratio when Dispersion moved by more than 0.25%**

Left axis- % vol terms from dispersion and V2X, Right axis- V2X ref



Source: J.P. Morgan Quantitative and Derivatives Strategy

In practice, one can implement a call spread collar structure on VSTOXX index to minimise decay and have delta on the spot. The amount of delta can be worked out by the beta of future to spot below. One can unwind the VSTOXX hedge and enter a new dispersion trade when the hedge makes money to enhance the dispersion entry level.

**Figure 39: V2X Future and Spot relationship from 5Y backtest**

|                            | 1M    | 2M    | 3M    |
|----------------------------|-------|-------|-------|
| 1 point change in V2X spot | 69.7% | 46.9% | 33.9% |
| No. future to hold         | 1.43  | 2.13  | 2.95  |

Source: J.P. Morgan Quantitative and Derivatives Strategy

### “Legging” the short index volatility position

Another technique to consider is “**index legging**” when implied volatility is very low. We tested a simple trading strategy on SX5E 1Y ATM volatility: we set a “target” improvement level for the volatility and a “stop-loss date” and we observe the index volatility every day and sell it if the target level is achieved. If the level is never reached over the observation window, we sell the index volatility anyway on the stop-loss date.

We found that **73% of the time** Euro STOXX 50 1Y volatility can be sold at least 20 bps higher within 15 days (more details can be found in the [previous report](#)). **Setting the target to 50 bps leads to a hit rate of 64% and the most likely PnL is 67 bps.** If we filter for when 1Y index vol to be below 18 at inception, we have a slight improvement on the hit ratio, but also a significant improvement on the average PnL as well as the max loss.

Figure 40: 0.5 vol target PnL for cases where 1Y index vol and V2X below 18 at inception

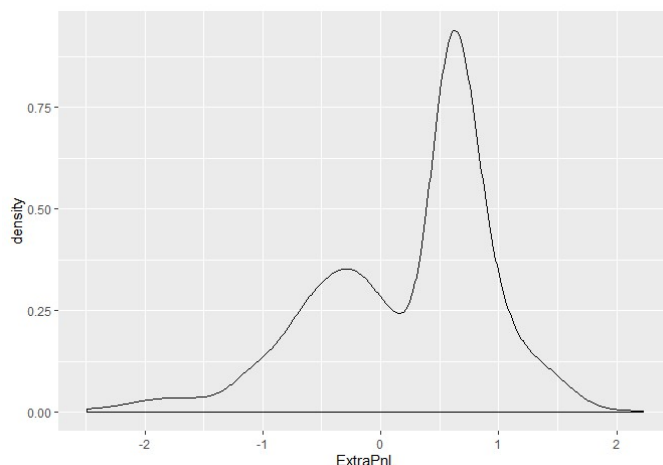
PnL in vol point

|                        | 15D         | 10D         |
|------------------------|-------------|-------------|
| <b>Target PnL</b>      | <b>0.5</b>  | <b>0.5</b>  |
| <b>Mode PnL</b>        | <b>0.63</b> | <b>0.62</b> |
| <b>%+ve trades</b>     | <b>65%</b>  | <b>59%</b>  |
| Mean PnL               | 0.24        | 0.17        |
| <b>Median PnL</b>      | <b>0.53</b> | <b>0.33</b> |
| Min PnL                | - 2.49      | - 2.29      |
| Max PnL                | 2.22        | 2.22        |
| % trade <stoploss date | 56%         | 45%         |
| % trade stopped        | 44%         | 55%         |

Source: J.P. Morgan Equity Derivatives Strategy.

Figure 41: Excess PnL distribution for 15D stop loss

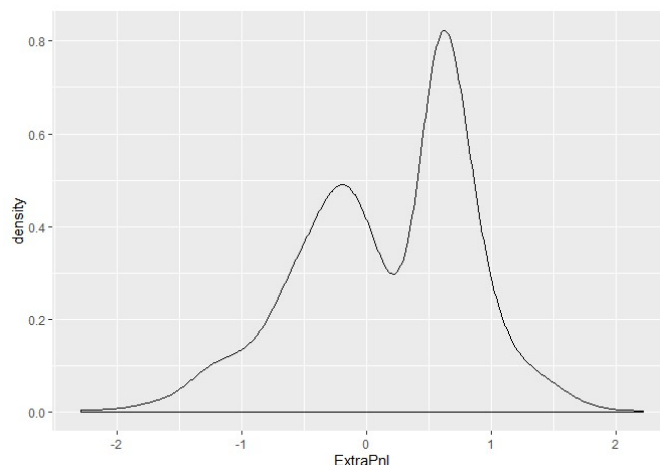
Conditional on 1Y index vol and V2X both below 18



Source: J.P. Morgan Equity Derivatives Strategy.

Figure 42: Excess PnL distribution for 10D stop loss

Conditional on 1Y index vol and V2X both below 18



Source: J.P. Morgan Equity Derivatives Strategy.

Therefore, when conditions suit, it is worth considering **buying a single stock leg only and selling index vol later** to achieve a better overall entry level for the package.

### Reverse dispersion overlay to improve carry

As dispersion is commonly expressed in vol swap format, risk is distributed across all the strikes on both legs. One way to improve carry during the life of the trade is **to sell the vol spread** to put on a “**reverse dispersion**” like position with positive premium.

The highest volatility spread is where implied correlation is the lowest. This is usually the **short tenor**, as correlation term structure is usually upward sloping as per the index vol. Similarly, the **strike is the upside** as single stock upside skew is much more flat versus the index.

One can hence enter a short-dated tenor upside strike (e.g. 3M110%) short stock long index position for a positive premium. We can often structure this with [lower strike on the index leg](#) to also have more **gamma protecting against correlation one moves**.

The stocks chosen to short should be “boring” stocks which either already had earnings or have low earnings sensitivity.

Figure 43: STOXX50 single stocks versus index volatility spread at different maturity and strikes

Ex-6names, forward vol spread

| Strike | 3M   | 1Y   | 2Y   |
|--------|------|------|------|
| 90%    | 6.3% | 6.5% | 6.6% |
| 100%   | 7.8% | 7.6% | 7.3% |
| 110%   | 8.9% | 8.5% | 7.9% |

Source: J.P. Morgan Equity Derivatives Strategy.

# Dispersion Levels Update

We provide the latest update of the vanilla volswap dispersion for Dec-24 expiry on the major European Indexes.

Figure 44: Indicative Offers for Volswap Dispersion

5Year Ptile is for the dispersion offer level vs historic entry level

|         | <i>index ref</i> | <i>YTD</i> | <i>Dec-24 offer</i> | <i>12M RVol spread</i> | <i>5Year Ptile</i> | <i>12M carry vs current offer</i> |
|---------|------------------|------------|---------------------|------------------------|--------------------|-----------------------------------|
| SX5E 44 | 4397.45          | 15.92%     | 8.70%               | 9.6%                   | 54.9%              | 0.9%                              |
| DAX T20 | 16041.37         | 15.21%     | 8.80%               | 9.4%                   | 76.0%              | 0.6%                              |
| SMI T10 | 11231.48         | 4.68%      | 6.90%               | 8.2%                   | 92.8%              | 1.3%                              |

Source: J.P. Morgan Equity Derivatives Strategy.

# Volatility Overview

## Corridor Variance and Skew Locks

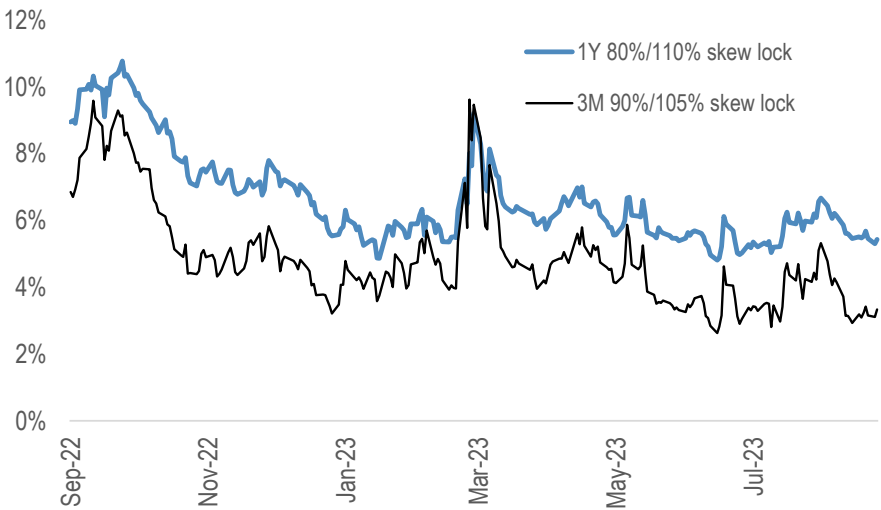
Figure 45: Euro STOXX 50 var vs. upvar and skew lock levels, 1Y percentile and weekly change

|    | Structure          | Current     | 1Y %ile | 1W change |
|----|--------------------|-------------|---------|-----------|
| 3M | 90% upvar          | 14.2%/15.0% | 17.5%   | 0.0%      |
|    | var                | 17.4%/18.2% | 15.1%   | 0.1%      |
|    | Var vs 90% upvar   | 2.4%/3.2%   | 5.1%    | 0.1%      |
|    | 90%/105% skew lock | 3.3%/4.1%   | 9.9%    | 0.2%      |
| 1Y | 80% upvar          | 16.4%/17.2% | 18.3%   | 0.0%      |
|    | var                | 21.3%/22.1% | 16.3%   | 0.0%      |
|    | Var vs 80% upvar   | 4.1%/4.9%   | 14.3%   | -0.1%     |
|    | 80%/110% skew lock | 5.4%/6.2%   | 12.3%   | 0.0%      |

Source: J.P. Morgan Equity Derivatives Strategy.

Figure 46: The Euro STOXX 50 skew lock trades over last year

Euro STOXX 50 downvar vs. upvar indicative levels (vol points)

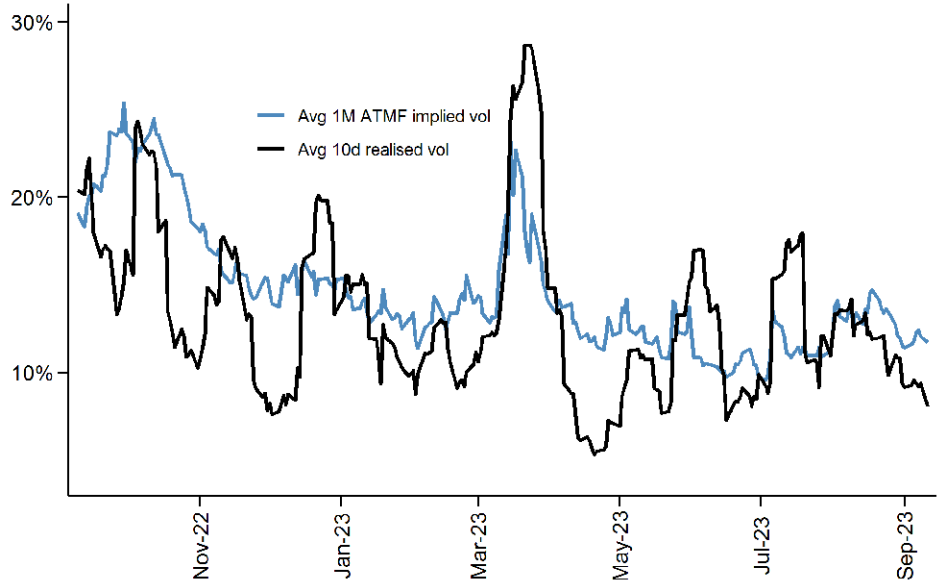


Source: J.P. Morgan Equity Derivatives Strategy.

## European Volatility

**Figure 47: European index short-dated implied and realized volatilities**

Avg. 1M ATMF and 10d realized volatility (SX5E, DAX, SMI and UKX)



Source: J.P. Morgan Equity Derivatives Strategy.

**Figure 48: Fixed-strike volatility 1W changes**

| Index   | Weekly Index Return | Sep-23 ATMF vol | Oct-23 ATMF vol | Sep-23 fixed-strike vol (1W chg) | Oct-23 fixed-strike vol (1W chg) |
|---------|---------------------|-----------------|-----------------|----------------------------------|----------------------------------|
| DAX     | 0.2%                | 12.1%           | 13.0%           | 0.2%                             | 0.0%                             |
| FTSEMIB | 0.4%                | 15.1%           | 15.2%           | 0.7%                             | 0.0%                             |
| SMI     | 0.7%                | 10.7%           | 10.3%           | 0.8%                             | -0.1%                            |
| SPX     | 0.2%                | 10.6%           | 11.3%           | 0.0%                             | -0.5%                            |
| SX5E    | 0.6%                | 13.4%           | 13.9%           | -0.0%                            | -0.2%                            |
| SX7E    | 2.0%                | 18.5%           | 19.5%           | 0.7%                             | -0.1%                            |
| UKX     | -0.6%               | 11.1%           | 10.8%           | 0.8%                             | 0.2%                             |

Source: J.P. Morgan Equity Derivatives Strategy.

## European Skew and Term Structure

Figure 49: 3M 25-delta skews, 5Y percentiles and 1Y range chart

| Index   | Current | 5Y %ile | 5Y Avg | 5Y Max | 5Y Min | 1W chg |   |
|---------|---------|---------|--------|--------|--------|--------|---|
| SX5E    | 4.5%    | 16%     | 7.0%   | 25.6%  | 3.3%   | 0.1%   | + |
| SPX     | 4.4%    | 1%      | 7.9%   | 25.1%  | 4.3%   | -0.1%  | + |
| UKX     | 3.1%    | 5%      | 5.7%   | 22.8%  | 2.7%   | 0.2%   | + |
| DAX     | 4.5%    | 16%     | 7.1%   | 22.0%  | 3.5%   | 0.1%   | + |
| SMI     | 2.7%    | 0%      | 5.1%   | 26.0%  | 2.5%   | 0.0%   | + |
| IBEX    | 3.9%    | 22%     | 5.9%   | 20.9%  | 2.5%   | 0.1%   | + |
| FTSEMIB | 4.2%    | 15%     | 6.8%   | 20.1%  | 3.4%   | -0.3%  | + |
| SX7E    | 5.0%    | 54%     | 5.5%   | 25.8%  | 1.4%   | 0.0%   | + |
| NKY     | 2.3%    | 3%      | 5.3%   | 23.1%  | 1.6%   | 0.1%   | + |
| HSI     | 2.0%    | 13%     | 3.7%   | 17.8%  | 0.5%   | 0.2%   | + |
| EEM.US  | 3.1%    | 2%      | 5.6%   | 21.3%  | 2.4%   | 0.4%   | + |

Source: J.P. Morgan Equity Derivatives Strategy.

Figure 50: 1Y less 3M ATM volatility term structures, 5Y percentiles and 1Y range chart

| Index   | Current | 5Y %ile | 5Y Avg | 5Y Max | 5Y Min | 1W chg |   |
|---------|---------|---------|--------|--------|--------|--------|---|
| SX5E    | 1.9%    | 78%     | 0.1%   | 3.2%   | -21.1% | -0.1%  | + |
| SPX     | 2.9%    | 85%     | 0.9%   | 4.6%   | -23.8% | -0.0%  | + |
| UKX     | 2.3%    | 85%     | 0.6%   | 3.4%   | -20.2% | 0.4%   | + |
| DAX     | 2.0%    | 85%     | 0.1%   | 3.3%   | -22.0% | -0.0%  | + |
| SMI     | 1.7%    | 75%     | 0.5%   | 2.9%   | -20.6% | -0.1%  | + |
| IBEX    | 1.4%    | 73%     | 0.0%   | 2.8%   | -22.0% | 0.3%   | + |
| FTSEMIB | 1.8%    | 78%     | -0.1%  | 3.2%   | -23.6% | 0.0%   | + |
| SX7E    | 2.6%    | 98%     | -1.3%  | 3.4%   | -26.2% | -0.8%  | + |
| NKY     | 1.8%    | 89%     | 0.3%   | 3.0%   | -18.2% | -0.0%  | + |
| HSI     | -0.3%   | 42%     | -0.2%  | 3.1%   | -13.7% | 0.3%   | + |
| EEM.US  | 3.0%    | 91%     | 1.0%   | 4.4%   | -21.9% | 0.2%   | + |

Source: J.P. Morgan Equity Derivatives Strategy.

## Call and Put Option Premia

Figure 51: Call option premia and 2Y percentiles

| Index   | 3M<br>105%F | 3M 105%F<br>%ile | 6M<br>105%F | 6M 105%F<br>%ile | 1Y<br>110%F | 1Y 110%F<br>%ile |
|---------|-------------|------------------|-------------|------------------|-------------|------------------|
| SX5E    | 0.8%        | 12%              | 1.8%        | 8%               | 2.1%        | 10%              |
| DAX     | 0.7%        | 7%               | 1.7%        | 7%               | 1.8%        | 7%               |
| SMI     | 0.4%        | 14%              | 1.2%        | 11%              | 1.3%        | 10%              |
| UKX     | 0.4%        | 13%              | 1.2%        | 9%               | 1.3%        | 12%              |
| SPX     | 0.5%        | 6%               | 1.5%        | 6%               | 1.6%        | 6%               |
| EEM     | 1.2%        | 10%              | 2.6%        | 11%              | 3.3%        | 22%              |
| SXAP    | 1.5%        | 1%               | 3.2%        | 2%               | 3.7%        | 0%               |
| SX7E    | 2.0%        | 5%               | 3.9%        | 7%               | 4.7%        | 8%               |
| FTSEMIB | 1.0%        | 10%              | 2.2%        | 7%               | 2.5%        | 8%               |
| IBEX    | 0.8%        | 9%               | 1.7%        | 3%               | 1.8%        | 2%               |
| CAC     | 0.7%        | 10%              | 1.7%        | 8%               | 1.7%        | 7%               |

Source: : J.P. Morgan Equity Derivatives Strategy.

Figure 52: Put option premia and 2Y percentiles

| Index   | 3M<br>95%F | 3M 95%F<br>%ile | 6M<br>90%F | 6M 90%F<br>%ile | 1Y<br>90%F | 1Y 90%F<br>%ile |
|---------|------------|-----------------|------------|-----------------|------------|-----------------|
| SX5E    | 1.3%       | 7%              | 1.6%       | 5%              | 3.2%       | 6%              |
| DAX     | 1.3%       | 4%              | 1.5%       | 3%              | 3.2%       | 6%              |
| SMI     | 0.8%       | 5%              | 0.9%       | 5%              | 2.1%       | 6%              |
| UKX     | 0.8%       | 2%              | 1.0%       | 6%              | 2.4%       | 10%             |
| SPX     | 1.1%       | 3%              | 1.4%       | 3%              | 3.2%       | 4%              |
| EEM     | 1.5%       | 6%              | 1.8%       | 8%              | 3.9%       | 11%             |
| SXAP    | 2.1%       | 2%              | 2.5%       | 5%              | 4.6%       | 2%              |
| SX7E    | 2.4%       | 7%              | 2.9%       | 7%              | 5.4%       | 11%             |
| FTSEMIB | 1.5%       | 6%              | 1.8%       | 4%              | 3.5%       | 6%              |
| IBEX    | 1.3%       | 2%              | 1.4%       | 2%              | 2.8%       | 2%              |
| CAC     | 1.2%       | 5%              | 1.4%       | 5%              | 3.0%       | 6%              |

Source: J.P. Morgan Equity Derivatives Strategy.

## Cross-Asset Derivatives

Table 2:

|        | Asset                   | Spot      | 1W Chg (%) | 1M Chg (%) | 3M ATM IV | 1W Chg (vol pt) | 1Y Low | 1Y High | 1Y Percentile | Current Relative to 1Y Low/High and 20th-80th Percentile Range | Z-Score | Implied-Realized | 1W Chg (vol pt) |
|--------|-------------------------|-----------|------------|------------|-----------|-----------------|--------|---------|---------------|----------------------------------------------------------------|---------|------------------|-----------------|
| Equity | Euro Stoxx 50           | 4,243.37  | -0.60      | -1.80      | 14.6%     | 0.08            | 12.7%  | 27.3%   | 18.3%         |                                                                | -0.82   | 0.74%            | 0.14            |
|        | FTSE                    | 7,536.25  | 1.32       | 0.16       | 11.5%     | -0.22           | 10.1%  | 22.3%   | 18.7%         |                                                                | -0.73   | 0.47%            | -0.21           |
|        | DAX                     | 15,733.56 | -0.24      | -0.62      | 14.1%     | 0.14            | 12.9%  | 28.3%   | 13.1%         |                                                                | -1.05   | 2.46%            | 0.43            |
|        | SMI                     | 11,021.73 | 0.57       | -0.54      | 11.4%     | 0.29            | 9.8%   | 20.4%   | 18.7%         |                                                                | -0.74   | 1.07%            | 0.15            |
|        | S&P 500                 | 4,487.46  | -0.63      | 0.52       | 12.9%     | 0.04            | 12.3%  | 28.2%   | 9.1%          |                                                                | -1.27   | 2.88%            | 0.29            |
|        | Nikkei 225              | 32,776.37 | -0.79      | 0.93       | 16.5%     | 0.49            | 15.0%  | 23.7%   | 25.6%         |                                                                | -0.69   | 1.32%            | 1.38            |
|        | Hang Seng               | 18,025.89 | -4.34      | -5.50      | 22.3%     | -0.09           | 19.6%  | 34.3%   | 44.8%         |                                                                | -0.50   | -0.03%           | -0.06           |
|        | H-Shares                | 6,260.19  | -4.18      | -4.29      | 25.3%     | 0.08            | 22.2%  | 38.3%   | 46.5%         |                                                                | -0.45   | -0.74%           | 0.41            |
| Comdty | KOSPI 200               | 335.54    | -1.40      | -1.01      | 13.3%     | 0.21            | 12.2%  | 22.4%   | 8.4%          |                                                                | -1.21   | 0.45%            | 0.49            |
|        | Copper                  | 376.30    | -1.26      | 1.31       | 18.8%     | 0.55            | 17.5%  | 31.4%   | 1.9%          |                                                                | -2.07   | 0.48%            | 0.17            |
|        | Gold                    | 1,919.10  | -0.37      | 0.32       | 10.5%     | -0.53           | 10.5%  | 18.5%   | 0.0%          |                                                                | -2.02   | 1.76%            | -0.53           |
| FX     | Crude Oil (WTI)         | 87.96     | 1.46       | 6.51       | 26.5%     | -1.68           | 25.8%  | 52.6%   | 0.3%          |                                                                | -1.68   | 2.10%            | 1.92            |
|        | EUR/USD                 | 1.07      | -0.41      | -1.95      | 6.8%      | 0.01            | 6.3%   | 12.8%   | 18.0%         |                                                                | -0.95   | 0.34%            | 0.11            |
| CDS    | GBP/USD                 | 1.25      | -0.71      | -1.36      | 8.0%      | 0.12            | 7.4%   | 18.8%   | 17.2%         |                                                                | -0.87   | 0.94%            | 0.43            |
|        | iTraxx Europe 5y        | 70.47     | 1.05       | -1.86      | 44.5%     | -0.24           | 38.9%  | 75.2%   | 11.4%         |                                                                | -0.96   | 11.13%           | -0.89           |
|        | CDX North America IG 5y | 63.24     | -0.86      | -4.84      | 40.8%     | -0.55           | 36.1%  | 64.6%   | 11.4%         |                                                                | -1.37   | 6.77%            | -0.76           |

|       | Asset   | Rate (%) | 1W Chg (spread) | 1M Chg (spread) | 3M ATM IV (bp vol) | 1W Chg (vol pt) | 1Y Low | 1Y High | 1Y Percentile | Current Relative to 1Y Low/High and 20th-80th Percentile Range | Z-Score |
|-------|---------|----------|-----------------|-----------------|--------------------|-----------------|--------|---------|---------------|----------------------------------------------------------------|---------|
| Rates | USD 10y | 4.00     | 0.02            | 0.07            | 109.7              | -0.08           | 100.0  | 156.0   | 23.3%         |                                                                | -0.83   |
|       | EUR 10y | 3.18     | 0.00            | -0.04           | 94.0               | -1.2            | 82.6   | 170.0   | 12.6%         |                                                                | -0.97   |

|              | Asset                       | Spread (bps) | 1W Chg (spread) | 1M Chg (spread) | Spread (bps) 1W Ago | Spread (bps) 1M Ago | 1Y Low | 1Y High | 1Y Percentile | Current Relative to 1Y Low/High and 20th-80th Percentile Range | Z-Score |
|--------------|-----------------------------|--------------|-----------------|-----------------|---------------------|---------------------|--------|---------|---------------|----------------------------------------------------------------|---------|
| CDS (spread) | iTraxx Europe 5y            | 70.47        | 0.73            | -1.34           | 69.7                | 71.8                | 66.8   | 138.2   | 9.1%          |                                                                | -0.96   |
|              | iTraxx x-Over 5y            | 395.24       | 2.75            | -9.09           | 392.5               | 404.3               | 377.6  | 662.3   | 11.1%         |                                                                | -0.86   |
|              | iTraxx Senior Financials 5Y | 81.31        | 1.19            | -0.59           | 80.1                | 81.9                | 77.0   | 152.2   | 9.1%          |                                                                | -0.99   |
|              | iTraxx Sub Financials 5Y    | 150.54       | 2.89            | 0.70            | 147.7               | 149.8               | 140.1  | 279.5   | 14.7%         |                                                                | -0.93   |
|              | CDX North America IG 5y     | 63.24        | -0.55           | -3.21           | 63.8                | 66.5                | 62.2   | 111.4   | 2.7%          |                                                                | -1.37   |
|              | CDX North America HY 5y     | 425.58       | -1.17           | -5.67           | 426.7               | 431.3               | 403.9  | 631.3   | 10.3%         |                                                                | -1.06   |

Table 3: Cross-asset correlations

6M weekly returns correlation

|           |                 | Equity        |      |      |         |            |           |           | Commodity |      |                 | FX   |      | Rates   |         | CDS           |               |                |                |           |           |
|-----------|-----------------|---------------|------|------|---------|------------|-----------|-----------|-----------|------|-----------------|------|------|---------|---------|---------------|---------------|----------------|----------------|-----------|-----------|
|           | Asset           | Euro Stoxx 50 | FTSE | DAX  | S&P 500 | Nikkei 225 | Hang Seng | KOSPI 200 | Copper    | Gold | Crude Oil (WTI) | EUR  | GBP  | USD 10y | EUR 10y | iTraxx Europe | iTraxx x-Over | iTraxx Snr Fin | iTraxx Sub Fin | CDX NA IG | CDX NA HY |
| Equity    | Euro Stoxx 50   |               | 82%  | 96%  | 71%     | 59%        | 64%       | 57%       | 39%       | -5%  | 44%             | 36%  | 17%  | 2%      | 10%     | -79%          | -80%          | -74%           | -73%           | -72%      | -73%      |
|           | FTSE            | 82%           |      | 83%  | 56%     | 55%        | 60%       | 54%       | 32%       | -13% | 57%             | 20%  | -13% | 4%      | 11%     | -77%          | -75%          | -72%           | -72%           | -62%      | -65%      |
|           | DAX             | 96%           | 83%  |      | 71%     | 64%        | 64%       | 57%       | 41%       | -5%  | 44%             | 32%  | 19%  | 5%      | 11%     | -78%          | -81%          | -73%           | -73%           | -69%      | -72%      |
|           | S&P 500         | 71%           | 56%  | 71%  |         | 51%        | 61%       | 57%       | 38%       | 17%  | 31%             | 31%  | 26%  | -7%     | -4%     | -59%          | -64%          | -49%           | -55%           | -75%      | -80%      |
|           | Nikkei 225      | 59%           | 55%  | 64%  | 51%     |            | 53%       | 60%       | 32%       | -22% | 23%             | -3%  | -11% | 22%     | 18%     | -56%          | -61%          | -54%           | -54%           | -45%      | -49%      |
|           | Hang Seng       | 64%           | 60%  | 64%  | 61%     | 53%        |           | 62%       | 51%       | 19%  | 39%             | 32%  | 24%  | -11%    | 0%      | -49%          | -53%          | -41%           | -37%           | -44%      | -48%      |
|           | KOSPI 200       | 57%           | 54%  | 57%  | 57%     | 60%        | 62%       |           | 43%       | 8%   | 34%             | 19%  | 4%   | 4%      | 6%      | -58%          | -61%          | -53%           | -52%           | -54%      | -57%      |
| Commodity | Copper          | 39%           | 32%  | 41%  | 38%     | 32%        | 51%       | 43%       |           | 20%  | 32%             | 39%  | 40%  | -1%     | 3%      | -43%          | -46%          | -40%           | -29%           | -35%      | -37%      |
|           | Gold            | -5%           | -13% | -5%  | 17%     | -22%       | 19%       | 8%        | 20%       |      | -19%            | 44%  | 54%  | -73%    | -59%    | 23%           | 24%           | 31%            | 42%            | 19%       | 18%       |
|           | Crude Oil (WTI) | 44%           | 57%  | 44%  | 31%     | 23%        | 39%       | 34%       | 32%       | -19% |                 | 11%  | -2%  | 18%     | 34%     | -58%          | -57%          | -58%           | -57%           | -45%      | -48%      |
| FX        | EUR             | 36%           | 20%  | 32%  | 31%     | -3%        | 32%       | 19%       | 39%       | 44%  | 11%             |      | 80%  | -48%    | -18%    | -22%          | -23%          | -17%           | -9%            | -10%      | -15%      |
|           | GBP             | 17%           | -13% | 19%  | 26%     | -11%       | 24%       | 4%        | 40%       | 54%  | -2%             | 80%  |      | -45%    | -23%    | 0%            | -4%           | 5%             | 13%            | 5%        | -2%       |
| Rates     | USD 10y         | 2%            | 4%   | 5%   | -7%     | 22%        | -11%      | 4%        | -1%       | -73% | 18%             | -48% | -45% |         | 82%     | -21%          | -23%          | -29%           | -38%           | -23%      | -23%      |
|           | EUR 10y         | 10%           | 11%  | 11%  | -4%     | 18%        | 0%        | 6%        | 3%        | -59% | 34%             | -18% | -23% | 82%     |         | -28%          | -30%          | -35%           | -38%           | -23%      | -24%      |
| CDS (5Y)  | iTraxx Europe   | -79%          | -77% | -78% | -59%    | -56%       | -49%      | -58%      | -43%      | 23%  | -58%            | -22% | 0%   | -21%    | -28%    |               | 98%           | 98%            | 91%            | 79%       | 79%       |
|           | iTraxx x-Over   | -80%          | -75% | -81% | -64%    | -61%       | -53%      | -61%      | -46%      | 24%  | -57%            | -23% | -4%  | -23%    | -30%    | 98%           |               | 95%            | 91%            | 83%       | 84%       |
|           | iTraxx Snr Fin  | -74%          | -72% | -73% | -49%    | -54%       | -41%      | -53%      | -40%      | 31%  | -58%            | -17% | 5%   | -29%    | -35%    | 98%           | 95%           |                | 92%            | 73%       | 72%       |
|           | iTraxx Sub Fin  | -73%          | -72% | -73% | -55%    | -54%       | -37%      | -52%      | -29%      | 42%  | -57%            | -9%  | 13%  | -38%    | -38%    | 91%           | 91%           | 92%            |                | 80%       | 80%       |
|           | CDX NA IG       | -72%          | -62% | -69% | -75%    | -45%       | -44%      | -54%      | -35%      | 19%  | -45%            | -10% | 5%   | -23%    | -23%    | 79%           | 83%           | 73%            | 80%            |           | 97%       |
|           | CDX NA HY       | -73%          | -65% | -72% | -80%    | -49%       | -48%      | -57%      | -37%      | 18%  | -48%            | -15% | -2%  | -23%    | -24%    | 79%           | 84%           | 72%            | 80%            | 97%       |           |

Source: J.P. Morgan Equity Derivatives Strategy.

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